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Name.....

Reg. No.....

THIRD SEMESTER (CBCSS–UG) DEGREE EXAMINATION, NOVEMBER 2020

Mathematics

MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

Answer at least **ten** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 30.

1. Find the derivative of $y = e^{-x\cos x}$.

2. Find the derivative of $y = \log\left(e^{2x} + e^{-2x}\right)$.

3. Find $\lim_{x \to 1^+} \frac{\sin \pi x}{\sqrt{x-1}}$.

4. Show that $\sinh^{-1} x = \log\left(x + \sqrt{x^2+1}\right)$.

5. Find $\lim_{n \to \infty} e^{\sin\left(\frac{1}{n}\right)}$.

6. Express $3.\overline{214}$ as a rational number.

7. Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{3+2^n}$ converges or diverges.

8. Show that the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}+1}$ is divergent.

9. Find a power series representation of $\frac{1}{(1-x)^2}$ on (–1, 1) by differentiating a power series representation of $\frac{1}{1-x}$.

10. Find the Maclaurin series of $\sin x$ and determine its interval of convergence.
11. Find a rectangular equation whose graph contains the curve C with the given parametric equation $x = 2t + 1, y = t - 3$.
12. Find the equation of the tangent line to the curve $x = \sec t, y = \tan t, \frac{-\pi}{2} < t < \frac{\pi}{2}$ at $t = \pi/4$.
13. Find the parametric equation for the line passing through the point (–2, 1, 3) and parallel to the vector $< 1, 2, -2)$.
14. The point $\left(3, \frac{\pi}{3}, \frac{\pi}{4}\right)$ is expressed in spherical co-ordinates. Find its rectangular co-ordinates.
15. Find the antiderivative of $r^1(t) = \cos t\, i + e^{-t} j + \sqrt{t}\, k$ satisfying the initial condition $r(0) = i + 2j + 3k$.

(10 × 3 = 30 marks)

Section B

Answer at least **five** *questions.*
Each question carries 6 *marks.*
All questions can be attended.
Overall Ceiling 30.

16. Use logarithmic differentiation to find the derivative of $y = \left(\sqrt{\cos x}\right)^x$.
17. Evaluate $\lim_{x \to 0} \frac{x^3}{x - \tan x}$.
18. Evaluate $\int_{-\infty}^{0} xe^x dx$.
19. Show that the series $\sum_{n=1}^{\infty} \left[\frac{2}{n(n+1)} - \frac{4}{3^n}\right]$ is convergent and find its sum.
20. Find the radius of convergence and interval of convergence of the series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 \cdot z^n}$.
21. Find the Taylor series for $f(x) = \log x$ at $x = 1$ and determine its interval of convergence.
22. Identify and sketch the surface $4x - 3y^2 - 12z^2 = 0$.
23. Let C be the helix $r(t) = 2\cos t\, i + 2\sin t\, j + t\, k, t \geq 0$. Find T (*t*) and N (*t*).

(5 × 6 = 30 marks)

Section C

Answer any **two** *questions.*
Each question carries 10 *marks.*

24. (a) Find $\int \frac{e^x}{e^{2x}+1}\,dx$.

(b) Find $\int \cosh(2x+3)\,dx$.

(c) Evaluate $\lim_{x \to 0} x^x$.

25. (a) Evaluate $\int_0^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$.

(b) Determine whether the series $\sum_{n=1}^{\infty} \frac{2n^2+n}{\sqrt{4n^7+3}}$ converges or diverges.

26. Find the area of the region that lies outside the circle $r = 3$ and inside the cardioid $r = 2 + 2\cos\theta$.

27. A shell is fired from a gun located on a hill 100 m above a level terrain. The muzzle speed of the gun is 500 m/sec and its angle of elevation is 30.

(a) Find the range of the shell.

(b) What is the maximum height attained by the shell ?

(c) What is the speed of the shell at impact ?

(2 × 10 = 20 marks